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(54) Title (EN): PRODUCING THE FZD7 CRD RECOMBINANT PROTEIN TO INHIBIT THE WNT SIGNALS AND CSCS WITH METHODS FOR IMPROVING THE SAME

(54) Title (FR): PRODUCTION DE LA PROTÉINE RECOMBINANTE CRD FZD7 POUR INHIBER LES SIGNAUX WNT ET LES CSCS AVEC DES PROCÉDÉS D'AMÉLIORATION ASSOCIÉS

(57) Abstract:

(EN): The present invention focuses on creating a soluble FZD7 receptor as a potential treatment for cancer. This new method aims to inhibit the Wnt/[3-catenin signaling pathway, which is crucial in controlling cancer cell behaviors like growth and survival, especially in cancer stem cells. CSCs are difficult to treat due to their ability to spread, return, and resist common treatments, so blocking this signaling pathway could effectively target these tough cells. The soluble FZD7 receptor, produced through recombinant technology, binds to Wnt ligands that typically activate the Wnt/[β-catenin pathway. By preventing these ligands from linking to FZD7 cell surface receptors, the soluble receptor can hinder Wnt signaling, thus reducing the formation of CSCs. This invention can be applied to treat various cancers, including breast, gastrointestinal, lung, and skin cancers, potentially improving treatment effectiveness and minimizing side effects. This method represents a promising advance in targeted cancer therapies.

(FR): La présente invention se concentre sur la création d'un récepteur FZD7 soluble en tant que traitement potentiel pour le cancer. Ce nouveau procédé vise à inhiber la voie de signalisation Wnt/[3-caténine, qui est cruciale pour lutter contre les comportements des cellules cancéreuses telles que la croissance et la survie, en particulier dans les cellules souches cancéreuses. Les CSC sont difficiles à traiter en raison de leur capacité à se propager, à se reconstituer et à résister à des traitements communs. Par conséquent, le blocage de cette voie de signalisation pourrait cibler efficacement ces cellules résistantes. Le récepteur FZD7 soluble, produit par la technologie recombinante, se lie à des ligands Wnt qui activent typiquement la voie Wnt/[β-caténine. En empêchant ces ligands de se lier aux récepteurs de surface cellulaire FZD7, le récepteur soluble peut bloquer la signalisation Wnt, réduisant ainsi la formation de CSC. La présente invention peut être appliquée pour traiter divers cancers, y compris les cancers du sein, gastro-intestinaux, du poumon et de la peau, améliorant potentiellement l'efficacité du traitement et réduisant au minimum les effets secondaires. Ce procédé représente une avancée prometteuse dans des thérapies anticancéreuses ciblées.

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